

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	Total Marks:	25
CO1 Statement:	<i>Apply advanced methodologies and agile project management techniques to manage software projects effectively</i>		
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Date:	22-07-2026	Duration:	60 minutes

Code of Conduct

I __A.Akshaya(192425170)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____ Akshaya_____

Instructions to Students:

ACADEMIC PROJECT, ALLOCATION MONITORING AND EVALUATION MANAGEMENT SYSTEM

Based on your allotted problem statement, prepare a report by answering the following questions:

1. Understand the Problem Statement

- Explain the problem in your own words.

The main problem is that many students are unable to work on projects independently and learn effectively from them. In many colleges, projects are often assigned based on students' CGPA or academic performance. It is commonly believed that students with a higher CGPA can perform better in projects. However, this is not always true because every student has unique talents, skills, and interests. Some students may have strong practical knowledge even if their CGPA is lower. Therefore, project allocation should not depend only on academic marks. Instead, it should consider each student's area of interest and technical skills. Students should be given projects in the domain they are passionate about, such as Web Development, Artificial Intelligence, Cyber Security, or Data Science. This will help them learn with greater interest and confidence. It also encourages students to improve the skills in which they feel they are weak. Beginners can gain knowledge step by step by working on projects they enjoy. Students will become more motivated to complete their projects successfully. This approach improves practical learning and innovation. It also helps students build better career opportunities in their chosen field. Hence, assigning projects based on students' interests and talents can improve both learning outcomes and project quality.

- Identify the objectives of the proposed system.

The main objective of the proposed system is to identify and understand each student's talents, interests, and technical skills before assigning a project. Instead of allocating projects only based on CGPA, the system focuses on the student's area of interest. If there is an academic application used by students, it should include a separate section called "**Student's Interest.**" In this section, students can select the domain they are interested in, such as Artificial Intelligence, Web Development, Cyber Security, Data Science, Cloud

Computing, or Mobile App Development. The system should also provide another section where students can select the programming languages they know or have completed learning, such as Java, Python, C++, HTML, CSS, or JavaScript. This information helps faculty members understand the strengths and preferences of each student. Based on these details, suitable projects can be allocated according to the student's skills and interests. This encourages students to work confidently in their preferred domain. It also helps beginners learn new technologies with greater interest. Faculty members can monitor students' progress more effectively. The system promotes fair and transparent project allocation. It improves practical learning, enhances technical skills, and prepares students for future career opportunities. Ultimately, the objective is to ensure that every student receives a project that matches their abilities, interests, and learning goals.

- State the scope of the problem.

The scope of this problem is very broad because it affects students' learning, project performance, academic results, and future careers. Many students do not get an opportunity to work in their area of interest because projects are often allocated without considering their skills and preferences. As a result, students may lose interest in their projects and fail to perform to their full potential. Every student has different talents, abilities, and career goals that should be recognized. If the system identifies each student's interests and technical skills before project allocation, suitable projects can be assigned accordingly. This allows students to work in domains they are passionate about, such as Artificial Intelligence, Web Development, Cyber Security, or Data Science. Students become more motivated to learn and complete their projects successfully. They can improve their practical knowledge and technical skills through hands-on experience. Faculty members can also provide better guidance because students will be working in their preferred domains. This leads to better project quality, improved academic performance, and increased confidence among students. It also helps students develop the skills required for placements and higher studies. Overall, allocating projects based on students' interests and talents creates a fair, effective, and career-oriented learning environment that benefits both students and educational institutions.



2. Identify Key Stakeholders and Challenges

- Identify all the stakeholders involved in the system.

The stakeholders involved in the **Academic Project Allocation, Monitoring and Evaluation Management System** include **Students, Faculty Guides, Project Coordinator, Head of Department (HOD), Management, System Administrator, External Examiner, and Placement Cell (optional)**. Students are the primary stakeholders because they use the system to select their interests, receive project allocations, submit progress reports, and view evaluation results. Faculty Guides mentor students, monitor their progress, and evaluate their work. The Project Coordinator manages project allocation, assigns guides, and monitors deadlines. The HOD supervises the overall project process and approves important decisions. The College Management monitors the overall project activities and reviews reports generated by the system. The System Administrator manages user accounts, maintains the database, and ensures the smooth functioning of the application. The External Examiner conducts the final evaluation and provides marks and feedback. The Placement Cell can use students' project information and skills to recommend suitable placement opportunities. All these stakeholders work together to ensure that projects are allocated fairly based on students' interests and skills, leading to better learning outcomes and career development.

- Explain the roles and responsibilities of each stakeholder.

Students are responsible for selecting their areas of interest, completing the assigned project, submitting progress reports, and presenting the final project.

Faculty Guides monitor students at every stage, provide technical guidance, help them correct errors, review their progress, and evaluate their performance based on the project output.

Project Coordinators are responsible for allocating projects, assigning faculty guides, setting deadlines, monitoring project progress, and ensuring that all students complete their work on time.

Head of the Department (HOD) supervises the entire project process, reviews students' project details, approves project allocations, and forwards them for further evaluation.

College Management oversees the overall functioning of the system, monitors project activities, and ensures that the project process is conducted smoothly.

System Administrator manages the application, creates user accounts, maintains the database, updates project information, and ensures the security and proper functioning of the system.

External Examiner conducts the final evaluation, reviews the project from all aspects, provides feedback, and awards the final marks that contribute to the student's academic project assessment. Each stakeholder has a specific responsibility, and together they ensure that projects are allocated fairly, monitored effectively, and evaluated accurately.

- Discuss the major challenges and issues faced by the stakeholders.

Every stakeholder involved in the project management system faces different challenges and responsibilities. No single stakeholder can be blamed because each person works under deadlines and expectations. **Students** may face coding errors, debugging difficulties, lack of technical knowledge, and stress due to project deadlines. Some students may also struggle to complete projects if they are not assigned to their area of interest. **Faculty Guides** experience pressure from the management to ensure that students complete their projects on time and maintain good project quality. The **Project Coordinator** has the responsibility of verifying projects, monitoring progress, assigning deadlines, coordinating with faculty members, and submitting marks and reports to the management within the scheduled time. The **Head of the Department (HOD)** monitors the overall project process and instructs faculty members to ensure timely completion of projects. The **College Management** expects all projects to be completed successfully before placement drives and academic evaluations. The **External Examiner** faces the challenge of fairly evaluating multiple projects and selecting the best ones based on quality, innovation, and performance.

Managing time, maintaining fairness, meeting deadlines, and ensuring high-quality project outcomes are common challenges faced by all stakeholders. Proper coordination and communication among all stakeholders are essential for the successful completion of the project process.

3. **Analyze the Current Approach**

- Describe the existing system or current process.

The existing system mainly focuses on attendance, internal marks, semester grades, and project evaluation. Projects are generally allocated based on academic performance, CGPA, or faculty decisions. Students usually do not get an opportunity to choose projects according to their interests or technical skills. Faculty members assign project guides and monitor students throughout the project period. Students submit progress reports at regular intervals and complete the project before the deadline. The project is then evaluated by faculty members and external examiners. The current process follows a fixed procedure for project allocation, monitoring, and evaluation. However, there is no proper mechanism to identify students' interests before assigning projects. Students with good practical skills may not receive projects related to their preferred domain. Some students lose interest because they are forced to work on unfamiliar topics. The system mainly measures academic performance rather than practical knowledge and creativity. As a result, students may not gain maximum learning from the project. Therefore, there is a need for a smarter system that considers students' interests and skills during project allocation.

- Identify its strengths and limitations.

Strengths

- It provides a structured process for project allocation and evaluation.
- Faculty members continuously guide students throughout the project.
- Students complete projects within fixed deadlines.
- Project evaluation is conducted in an organized manner.
- Academic records, marks, and reports are maintained properly.

Limitations

- Projects are not allocated based on students' interests or talents.

- Students may lose interest if they receive projects outside their preferred domain.
 - The system mainly considers academic performance instead of practical skills.
 - Students with lower CGPA but good technical skills may not get suitable opportunities.
 - There is no feature to identify students' coding knowledge or preferred programming languages before project allocation.
- Analyze the problems or gaps that need to be addressed.

The biggest gap in the current system is that students' interests, talents, and technical skills are not considered while allocating projects. Every student has different strengths and career goals. Some students may be interested in Web Development, while others may prefer Artificial Intelligence, Cyber Security, Data Science, Cloud Computing, or Mobile App Development. The current system does not provide an option for students to mention their interests or programming skills. It also does not check whether students have completed learning programming languages such as Java, Python, C++, HTML, CSS, or JavaScript. Because of this, many students are assigned projects that do not match their abilities. This affects their confidence, learning experience, and project performance. The system should allow students to select their preferred domain and technical skills before project allocation. Projects should then be assigned based on this information. This will improve students' practical knowledge, increase their interest in learning, and help them perform better in their academic projects and future careers.

4. **Propose a Suitable Solution Using Software Engineering Principles**

- Design a software-based solution for the given problem.

The proposed solution is an **Academic Project Allocation, Monitoring and Evaluation Management System**. The main purpose of this system is to allocate projects based on students' interests, talents, and technical skills instead of only considering their CGPA. Every student will create an account and complete their profile. The profile includes personal details, preferred project domain, programming languages known, and technical skills. Students can choose domains such as Artificial Intelligence, Web Development, Cyber Security, Data Science, Cloud Computing, or Mobile App Development. Based on these details, the system recommends suitable projects. Faculty members can review the recommendations and approve the project allocation. Students can upload project reports, documents, and weekly progress updates through the system. Faculty members can monitor

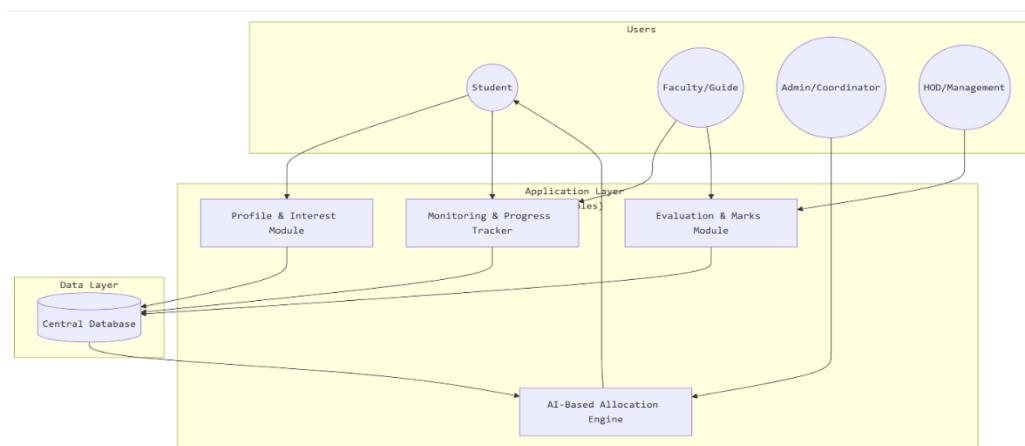
progress, provide suggestions, and evaluate the work regularly. The system also sends reminders about deadlines and project reviews. HOD, Project Coordinator, and Management can monitor the overall project status. External examiners can evaluate the final project and submit marks. This system ensures fair project allocation and helps students improve their practical knowledge and technical skills.

- Describe the major functional modules of the proposed system.

Student Module allows students to register, log in, update their profile, select their interests, choose programming languages, and view allocated projects.

Project Allocation Module matches students' interests and technical skills with available projects and assigns faculty guides.

Faculty Module enables faculty members to monitor students, review progress, provide feedback, and evaluate project performance.



The **Project Monitoring Module** helps students upload weekly reports and allows faculty to track project completion.

Evaluation Module records internal assessments, external evaluations, marks, and feedback.

Project Coordinator Module manages project allocation, deadlines, and progress reports.

HOD Module supervises project activities and approves project-related decisions.

Management Module generates reports and monitors overall project completion.

Admin Module manages user accounts, project information, departments, and system maintenance. Together, these modules ensure smooth project allocation, monitoring, and evaluation.

- Identify the functional and non-functional requirements.

The functional requirements include student registration and login, faculty login, HOD login, and administrator login. Students should be able to enter their interests, technical skills, and preferred programming languages. The system should allocate projects based on students' profiles. Students should upload project documents and progress reports. Faculty members should review submissions and provide feedback. The system should allow project coordinators to manage deadlines and monitor project progress. HOD and Management should be able to view project reports and statistics. External examiners should evaluate projects and submit marks. Notifications should be sent for project reviews and deadlines. The non-functional requirements include a user-friendly interface that is easy to understand. The system should provide high security for student and project data. It should be reliable and available whenever users need it. The software should be scalable to support more students and departments. It should also be easy to maintain, update, and provide good performance even when many users access it simultaneously.

- Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.

Software engineering principles play an important role in developing this system. **Modularity** is achieved by dividing the system into separate modules such as Student, Faculty, Project Allocation, Monitoring, Evaluation, and Administration. **Scalability** allows the system to support more students, departments, and projects in the future without major changes. **Maintainability** ensures that individual modules can be updated or modified without affecting the entire system. **Reliability** ensures accurate project allocation, secure storage of data, and proper monitoring of project progress. **Usability** provides a simple and user-friendly interface that makes the system easy for students, faculty, and administrators to use. **Security** is maintained through separate login

credentials and role-based access for students, faculty, HOD, project coordinators, examiners, and administrators. The system also protects confidential project information and student records. Regular backups and validation of user data improve system safety and performance. These software engineering principles make the system efficient, flexible, secure, and easy to manage for all stakeholders.

5. Justify the Chosen Methodology/Framework

- Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).

The **Agile Software Development Life Cycle (SDLC) Model** is selected for developing the **Academic Project Allocation, Monitoring and Evaluation Management System**. Agile is one of the most popular software development methodologies because it is flexible and supports continuous improvement. The system requirements may change based on feedback from students, faculty members, HOD, and management. Agile allows developers to divide the project into small parts called iterations or sprints. Each iteration focuses on developing and testing a specific feature of the system. For example, the Student Module, Project Allocation Module, Monitoring Module, and Evaluation Module can be developed separately. After completing each module, users can test it and provide feedback. Any errors or changes can be corrected before moving to the next iteration. This reduces development risks and improves software quality. Agile encourages communication between developers and stakeholders throughout the project. It also ensures that the final system meets user requirements effectively.

- Justify why the selected methodology/framework is suitable for the proposed system.

Agile is the most suitable methodology for this system because the project requirements may change during development. Educational institutions may introduce new project domains, evaluation methods, or monitoring features at any time. Agile allows these changes to be implemented without affecting the entire system. It supports continuous communication between developers and stakeholders, including students, faculty members, project coordinators, HOD, and management. Feedback collected after each development phase helps improve the software. Testing is performed regularly, making it easier to identify and fix errors early. The modular development approach allows different system features to be completed independently. This saves both time and development cost. Agile also improves software quality because users can review each feature before the final

release. Since this system requires frequent updates and user feedback, Agile provides flexibility, faster development, and better project management than traditional SDLC models.

- Explain how it supports successful project development and maintenance.

The Agile methodology supports successful project development by dividing the work into smaller and manageable stages. Each stage is planned, developed, tested, and reviewed before moving to the next one. This helps developers identify problems at an early stage and correct them immediately. Regular meetings and discussions with stakeholders ensure that the system meets user requirements. Continuous testing improves software quality and reduces the chances of major errors. If students, faculty members, or management request new features, they can be added in future iterations without redesigning the entire system. Agile also makes software maintenance easier because each module is developed independently. Developers can update one module without affecting the others. Regular feedback helps improve system performance and user satisfaction. The methodology also supports quick bug fixing and feature enhancement after deployment. As a result, the system remains reliable, secure, flexible, and easy to maintain throughout its lifecycle.

6. Discuss Expected Outcomes and Limitations

- Describe the expected benefits of the proposed solution.

The proposed **Academic Project Allocation, Monitoring and Evaluation Management System** provides a better approach for managing student projects. The main aim of this system is to identify students' interests, talents, and technical skills before assigning projects. It helps students work on projects that match their abilities instead of only depending on CGPA.

The major benefits of the proposed system are:

- Projects are allocated based on students' interests and skills.
- Students get opportunities to work in their preferred domains.
- It improves students' practical knowledge and technical abilities.
- Faculty members can monitor student progress easily.
- Project allocation becomes fair and transparent.
- Management can track project status and generate reports.

This system increases students' confidence and motivation because they work on projects they are interested in. It also helps students improve their problem-solving skills and prepare for future career opportunities. The system creates a better learning environment where every student gets an equal chance to showcase their talent.

- Explain how the solution addresses the identified challenges.

The current system mainly focuses on CGPA and academic marks while allocating projects. The proposed solution solves this problem by considering students' personal interests, programming skills, and preferred domains. It ensures that students are not forced to work on topics where they have less knowledge or interest.

The system addresses challenges by:

- Allowing students to select their domain of interest.
- Collecting information about programming languages known by students.
- Recommending suitable projects based on skills.
- Providing continuous monitoring and feedback.
- Reducing project-related stress among students.
- Improving communication between students and faculty.

Faculty members can provide better guidance because they understand students' abilities and project requirements. Project coordinators can manage deadlines effectively, and management can easily monitor overall project development. This improves project quality and helps students gain meaningful practical experience.

- Discuss potential risks, implementation challenges, and limitations.

Although the proposed system provides many advantages, some risks and challenges may occur during implementation. Students may enter incorrect information about their skills and interests, which can affect project allocation. Some students may not be clear about their preferred domain or career goals. Faculty members and administrators may require training to use the new system effectively. Developing and maintaining the software requires time, cost, and technical resources. Data security is another important challenge because the system stores student details, project information, and evaluation records. Internet connectivity issues may affect online submissions and monitoring. The system

may require regular updates to include new technologies and project domains. Initially, users may face difficulty adapting to the new system. The accuracy of project recommendations depends on the information provided by students. The system may also require continuous maintenance and technical support. These limitations can be reduced through proper planning, user training, and regular system improvements.

- Suggest possible future enhancements.

The proposed system can be improved further by adding advanced features in the future. An AI-based recommendation system can be introduced to automatically suggest projects based on students' interests, skills, and previous performance. A skill assessment test can be added to understand students' technical abilities before project allocation. Machine Learning techniques can be used to improve project recommendations. The system can be integrated with platforms like GitHub to monitor coding activities and project development. A mobile application can be developed for easy access by students and faculty members. Automated notifications through email and mobile alerts can remind students about deadlines and reviews. A plagiarism detection feature can be added to check project originality. Analytics dashboards can help management analyze student performance and project outcomes. Industry mentors can be included to provide real-world guidance. The system can also include career recommendations based on students' project domains and skills. These enhancements can make the system more intelligent, efficient, and useful for future educational requirements.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis	Identifies all stakeholders, their roles, and	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are

	challenges with clear analysis.			incomplete or inaccurate.
Current System Analysis	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.
Proposed Software Solution & Software Engineering Principles	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability, maintainability, security, etc.).	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.
Methodology/ Framework Justification	Selects the most suitable SDLC/model/framework and provides strong technical justification.	Suitable methodology selected with adequate justification.	Methodology selected but justification is weak.	Outcomes, limitations, or future enhancements are missing; report lacks organization and clarity.

Criteria	Mark
Problem Understanding	/5
Stakeholder & Challenge Analysis	/5
Current System Analysis	/5
Proposed Software Solution & Software Engineering Principles	/5
Methodology/ Framework Justification	/5
TOTAL	/5